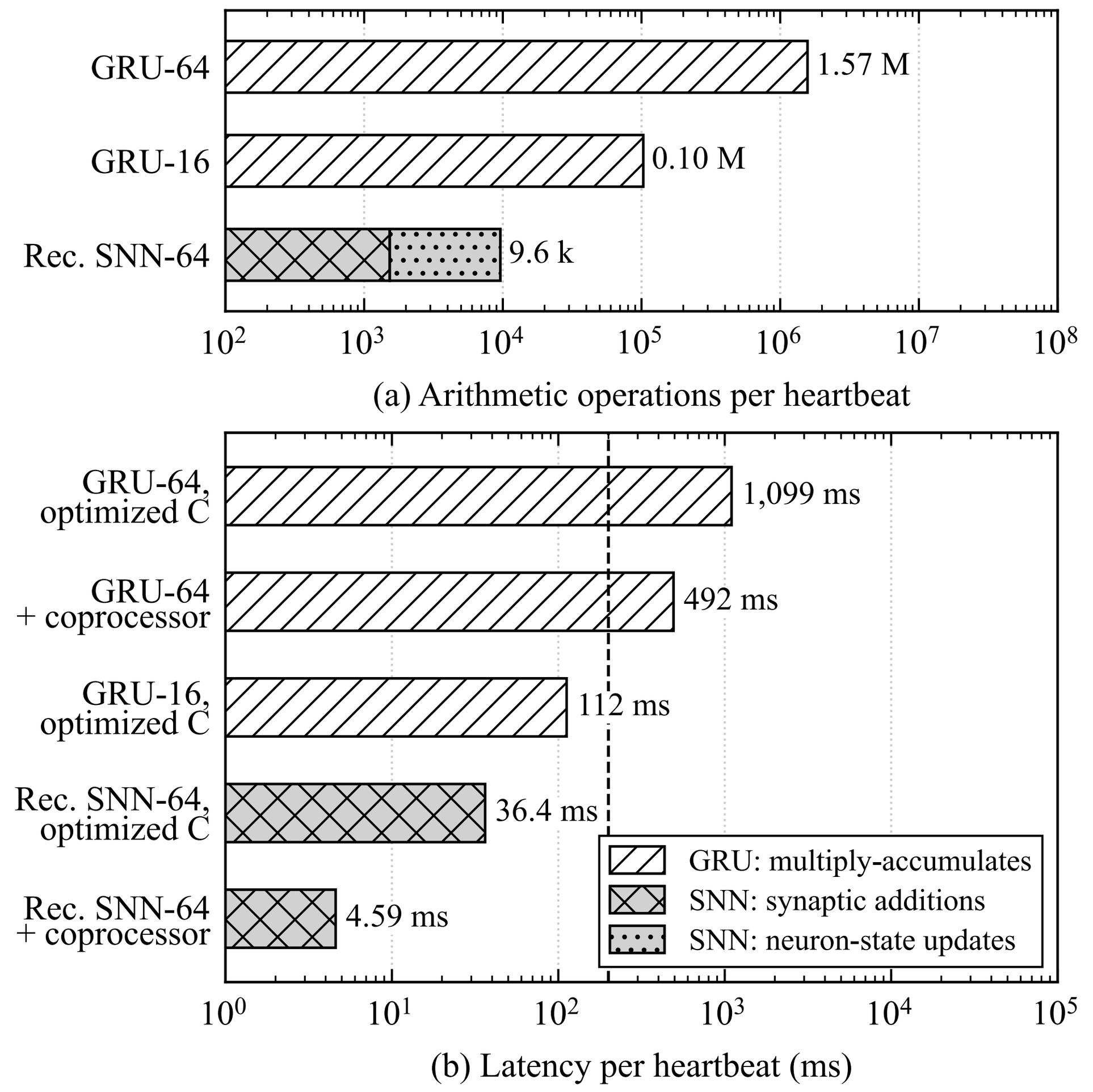

GRU-64
1.57 M
GRU-16
0.10 M
Rec. SNN-64
9.6 k
10^2
10^3
10^4
10^5
10^6
10^7
10^8
(a) Arithmetic operations per heartbeat
GRU-64, optimized C
1,099 ms
GRU-64 + coprocessor
492 ms
GRU-16, optimized C
112 ms
Rec. SNN-64, optimized C
36.4 ms
Rec. SNN-64 + coprocessor
4.59 ms
GRU: multiply-accumulates
SNN: synaptic additions
SNN: neuron-state updates
10^0
10^1
10^2
10^3
10^4
10^5
(b) Latency per heartbeat (ms)